

When high flux helps single-pixel imaging: a contrast–dead-time operating map

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Abstract

Single-pixel imaging with photon-counting detectors is conventionally run at low flux so that detector dead time removes few arrivals, but this protection costs acquisition time. We ask when deliberately entering the nonlinear dead-time regime improves count-only single-pixel imaging and when it merely adds saturated photons.

A convex renewal quasi-likelihood (RQL) reconstruction models the integrated nonparalyzable count without per-scene tuning. Its finite-window Fisher information has a principal ridge at $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, while the bucket-energy contrast of the illumination patterns determines how much of that scalar count information reaches scene structure.

In a preregistered dense-pattern study, the time-to-quality endpoint was uninformative and therefore formally negative for every scene; a small but uniformly positive fixed-dwell benefit remained, with median gain +0.28 dB. Dense multiplexing thus suppressed structural modulation even when dead-time correction remained accurate.

In a second preregistered study, spatial occupancy varied at equal mean detector load and equal incident dose across occupancies. The acquisition-speed primary remained formally negative because its breadth criterion failed, whereas the fixed-dwell secondary passed with a median +1.16 dB gain. The response across occupancies was nonmonotone, exposing a contrast–conditioning trade-off rather than a universal optimal sparsity.

Together, the results define a contrast–dead-time operating map: high flux is useful only when illumination preserves enough bucket modulation for the scene and the operator remains sufficiently conditioned. Sparse equal-dose operation also concentrates power into fewer illuminated pixels, so peak irradiance can bind before the detector limit. All evidence here is simulation-only and requires optical-bench validation.

1 Introduction

Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the temporal sequence

of a single bucket detector’s responses to structured illumination [1, 2, 3], and have proven particularly attractive where array sensors are unavailable or photon budgets are severe: low-light and photon-starved imaging, NIR/SWIR sensing, and imaging through scattering media such as turbid water [4, 5, 6]. In the photon-starved regime the bucket detector is a photon-counting device — a single-photon avalanche diode or photomultiplier — and image quality is set by how many photons can be counted within the available acquisition time.

Photon-counting detectors, however, are blind for a dead time τ after each registered count. The established response is to operate at low flux — keeping the dead-time duty $\rho = \lambda\tau$ at the few-percent level so that losses stay small and a linear correction suffices — and to pay for it with acquisition time, since the count rate is then a small fraction of the detector’s $1/\tau$ ceiling. A substantial literature instead corrects or models the distortion at higher rates: dead-time compensation and high-flux operation in single-photon lidar [7, 8, 9, 10], count-rate corrections in photon-counting imaging [11], and dead-time-aware estimation bounds [12]. These works are correction-centric or timestamp-based; what remains unmapped is the operating question for the count-only bucket channel itself: at what flux, for which scenes, does raising the rate into the nonlinear regime actually help?

Previous studies established count-rate-dependent CRLB degradation, task-specific optimal operating fluxes, and high-flux recovery using time-resolved dead-time models [13, 14, 9, 7, 12]. Here, we study a different information bottleneck: a finite exposure from which only the integrated nonparalyzable count is retained. For this count-only observation, we derive a principal finite-window Fisher-information ridge and show that its optimal load obeys a cube-root law, $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, arising from terminal detector-phase information loss.

Sparse illumination itself is not the claimed contribution of this work. Prior work has used sparse patterns to reduce detector count rates and alleviate pile-up [15, 16]. Here, instead, spatial occupancy is varied under equal mean detector load and equal incident dose, revealing a contrast-controlled boundary between multiplex-limited and photon-limited high-flux single-

pixel imaging.

This work contributes: (i) an exact finite-window count-information analysis of the nonparalyzable channel, including a missing-information identity and the cube-root ridge of Eq. (5); (ii) a convex renewal quasi-likelihood reconstruction with a truth-free analytic regularization rule, validated against the exact likelihood; (iii) two preregistered simulation studies — dense-pattern and equal-load occupancy-ladder — whose endpoints, gates, scene lists, and display subjects were frozen in immutable commits before any confirmatory reconstruction existed, and whose outcomes are reported as frozen, including the formally negative primaries; and (iv) the resulting operating map with its computable engagement criterion. Figure 1 sets out the single-pixel imaging chain, the dead-time saturation mechanism, and the resulting contrast–dead-time operating map. All code, frozen preregistration documents, and per-cell evidence bundles are openly released.

2 Physical model and problem statement

A single-pixel detector with dead time τ observes, for illumination pattern a_i and scene $x \geq 0$, a Poisson arrival process of rate $\lambda_i = \Phi a_i^\top x + d$ during an exposure of duration $T = \nu\tau$; after each registered count the detector is blind for τ (non-paralyzable, active start). The recorded observation is the integrated count N_i alone, with exact law

$$\begin{aligned} P(N_i \geq m) &= F_{\Gamma(m, \lambda_i)}(T - (m-1)\tau) \\ &= P(\text{Pois}(z_m) \geq m), \\ z_m &= \lambda_i(T - (m-1)\tau). \end{aligned} \quad (1)$$

The per-slot load $\rho_i = \lambda_i\tau$ is the dead-time duty; the conventional regime keeps $\rho \lesssim 0.05$. Illumination ensembles used in this work: dense Bernoulli(1/2) patterns (Study 1 primary), complementary Hadamard-type pairs charged at $2M$ physical exposures and gamma-modulated patterns (supplementary ensemble sensitivity), and the Study-2 fixed-occupancy binary ladder of Section 6. Because patterns spread instantaneous load, deployment statements use the per-pattern 95th percentile of ρ rather than its mean; percentile ledgers for every cell are part of the released evidence bundles.

3 Renewal quasi-likelihood reconstruction

We reconstruct by

$$\begin{aligned} \hat{x} = \arg \min_{x \geq 0} \frac{1}{M} \sum_{i=1}^M &\left[(T - N_i\tau)\lambda_i(x) - N_i \log \lambda_i(x) \right] \\ &+ \lambda_{\text{TV}} \text{TV}(x), \end{aligned} \quad (2)$$

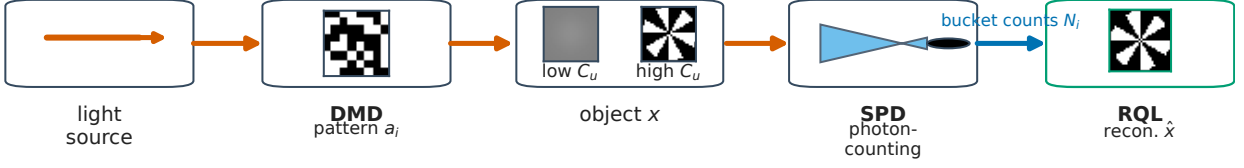
with $\lambda_i(x) = \Phi a_i^\top x + d$. The data term is convex in λ (and in x through the affine map); its single-measurement stationary point $\hat{\lambda} = N/(T - N\tau)$ coincides with the classical dead-time rate correction, so the estimator’s novelty resides not in the per-measurement formula but in the joint regularized inverse problem and the operating-point map it enables. At ceiling counts ($N\tau \geq T$) the objective is unbounded below in λ — the true identifiability boundary of count-only observation; the fraction of ceiling counts is recorded per cell. A full heteroscedastic Gaussian likelihood fails at high load through a variance-collapse non-coercivity (Supplement §S1).

3.1 Truth-free regularization selection and descriptive audit

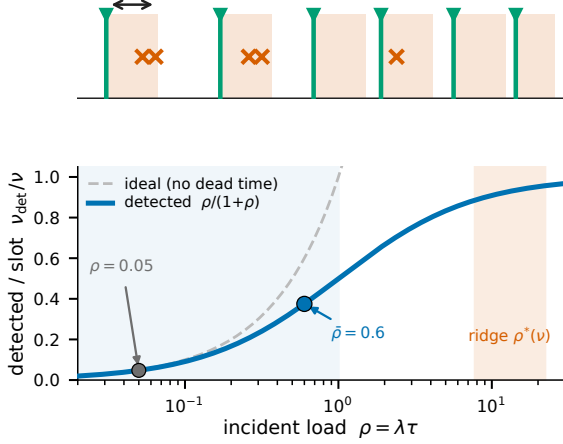
The TV weight is set by a frozen analytic score-concentration rule with no per-scene tuning: $\lambda_{\text{TV}} = c\sigma_g\sqrt{2\ln n}$ with global constant $c = 0.50$ frozen before any confirmatory data existed, and $\sigma_g = \Phi\sqrt{\kappa_A v_s/M}$ computed from the exact renewal per-count score variance v_s (no Gaussian approximation; the central limit theorem is quantitatively untrustworthy at short windows $\nu \leq 10$). All arms are scaled by the identical rule through their own κ_A .

The measurement audit is descriptive and does not constitute a model-adequacy test. All audit statistics are excluded from regularization selection, reconstruction, cell inclusion, and confirmatory pass/fail decisions. The refit-bootstrap reference distribution is conditional on a smoothed plug-in scene; its empirical tail ranks are therefore not calibrated model-class p -values for arbitrary scenes and can be anti-conservative in the upper tail when the true scene is rougher than the plug-in. Moreover, count-only observations collected at a single operating point cannot generally distinguish detector mean-response mismatch from radiometric rescaling of the scene. Detector-mismatch consequences are therefore quantified directly through the preregistered radiometric metrics in Section 6.6.

(a)



(b)



(c)

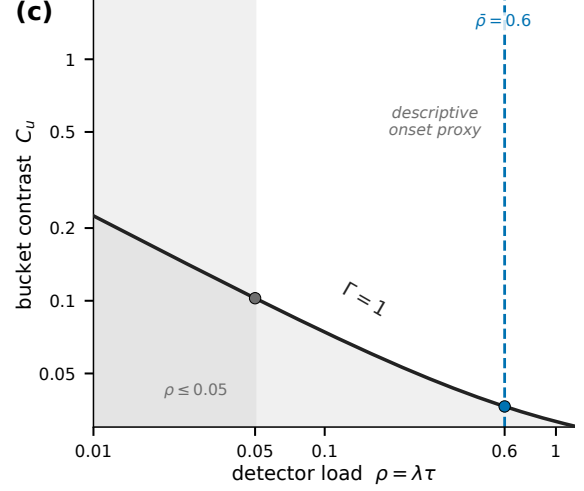


Figure 1: **High-flux single-pixel imaging is controlled by detector load and bucket contrast.** (a) Computational acquisition model: a global flux scale Φ drives DMD patterns a_i on a scene x ; the bucket counts N_i feed renewal quasi-likelihood (RQL) reconstruction $\hat{x}$ under $\lambda_i = \Phi a_i^\top x + d$ and exposure $T = \nu \tau$. (b) After each registered count the detector is blind for τ , censoring later arrivals. The mean detected counts bend away from the ideal response, while exact count information peaks at $\rho^*(\nu) = (6\nu)^{1/3} - 2/3$ ($\rho^* \approx 7.8$ and 22.2 at $\nu = 100$ and 2000), beyond the conventional $\rho = 0.05$ and tested high-flux $\bar{\rho} = 0.6$ points. (c) The onset $\Gamma = 1$ in the contrast–load plane is a descriptive photon-limited-onset proxy, not a predicted or validated boundary. Grayscale object icons are illustrative.

4 The finite-window count-information ridge

4.1 The missing-information identity

Let $I_N(\rho, \nu)$ denote the Fisher information of the single-frame count N with respect to $\log \lambda$, and $J = I_N / \nu$ its per-slot normalization ($J = 1$ is the saturation ceiling: one unit of log-rate information per dead-time slot). Writing $R_\nu \in [0, 1)$ for the residual dead-time phase left at the window end — the unobserved fraction of a dead-time period truncated by the frame boundary — the information splits as

$$I_N = \mathbb{E}[N] - \rho^2 \mathbb{E}[\text{Var}(R_\nu | N)]. \quad (3)$$

The first term is the complete-data information of the underlying renewal record (each registered event contributes one unit on the $\log \lambda$ scale); the second is the price of not observing the terminal phase: the frame boundary censors the last dead-time period, and the count alone cannot reveal how much of the window was spent in that final blind interval. The terminal phase has $P(R_\nu = 0) = 1/(1 + \rho)$ and density $\rho/(1 + \rho)$ on $(0, 1)$, so $\text{Var}(R_\nu) \rightarrow \rho(\rho + 4)/[12(1 + \rho)^2] \rightarrow 1/12$ at high load — the censoring price grows as $\rho^2/(12\nu)$ per slot.

4.2 The cube-root law

Expanding J at large ρ and large ν ,

$$J(\rho, \nu) = 1 - \frac{1}{\rho} - \frac{\rho^2}{12\nu} + \frac{1}{\rho^2} - \frac{\rho}{6\nu} + \frac{\rho^4}{144\nu^2} + o(\nu^{-2/3}). \quad (4)$$

The leading tradeoff is between the saturation gap $1/\rho$ (a slot only carries full rate information when occupied by a completed detection–dead-time cycle) and the censoring price $\rho^2/(12\nu)$. The marginal balance $1/\rho^2 = \rho/(6\nu)$ gives $\rho^3 = 6\nu$, i.e. the information-optimal load

$$\rho^*(\nu) = (6\nu)^{1/3} - \frac{2}{3} + O(\nu^{-1/3}), \quad (5)$$

with peak per-slot information

$$\begin{aligned} J(\rho^*) &= 1 - (c^{-1} + c^2/12)\nu^{-1/3} + o(\nu^{-1/3}) \\ &= 1 - 0.8255\nu^{-1/3} + o(\nu^{-1/3}), \quad c = 6^{1/3}. \end{aligned} \quad (6)$$

Exact evaluation of the renewal PMF gives $\rho^* = 4.53, 6.16, 7.87, 9.99, 13.77, 17.45$, and 22.16 at $\nu = 20, 50, 100, 200, 500, 1000$, and 2000 , respectively; the closed form $(6\nu)^{1/3} - 2/3$ gives $4.27, 6.03, 7.77, 9.96, 13.76, 17.50$, and 22.23 — a 5.9% gap at $\nu = 20$ shrinking to sub-percent by $\nu = 500$, consistent with the $O(\nu^{-1/3})$ remainder [Fig. 3(a)].

Two structural remarks. (i) The ridge is not a count-variance pinning effect: at $\rho = \rho^*$ the count variance $\text{Var } N = \nu^{1/3}/6^{2/3}$ still diverges with ν ; counts pin to the ceiling only at the much larger scale $\rho \asymp \sqrt{\nu}$. The information peak is an information-geometric tradeoff, not a saturation artifact. (ii) Beyond the ridge ($\rho \geq \rho^*$) the count information genuinely decreases — no estimator operating on counts, including the exact-likelihood reference, recovers it. High-flux operation is therefore beneficial only up to a computable boundary, and that boundary is one axis of the operating map reported here.

4.3 Deployment zoning and the CLT trust boundary

The Gaussian (CLT) surrogate information satisfies $J_{\text{exact}}/J_{\text{CLT}} \simeq 1 - \rho^2/(12\nu)$, so the surrogate stays within 10% of exact only for $\rho \leq \rho_{0.9} \asymp \sqrt{1.2\nu}$ — a $\sqrt{\nu}$ scaling, distinct from the $\nu^{1/3}$ ridge. We partition the (ρ, ν) plane into an RQL deployment zone (per-pattern $\rho_{95} \leq 1$, the regime the campaigns demonstrate), a transition band, an exact-reference mode (short-window and extreme-saturation reference computations with the exact PMF), and the information-decreasing region $\rho \geq \rho^*(\nu)$. Zoning uses the per-pattern 95th percentile of ρ , not its mean: spread-occupancy patterns distribute per-frame load, and the deployment guarantee must hold for the loaded tail, not the average.

4.4 Relation to prior work

Müller systematized renewal-theoretic analysis of extended and nonextended dead-time counters, relating distorted interarrival laws to the resulting count statistics and measurement-origin effects [17]. Yu and Fessler derived exact first and second moments for several dead-time counting models and used those moments to study intensity correction and the adequacy of Poisson reconstruction models [18]. Wang et al. derived pileup count statistics for idealized energy-discriminating nonparalyzable x-ray detectors and evaluated task-specific Cramér–Rao performance as a function of input rate, establishing substantial high-rate degradation of spectral-counting advantages [13]. Alvarez used renewal-process large-count moments and multivariate-Gaussian count models to quantify pileup-induced covariance, signal-to-noise loss, and material-estimation Cramér–Rao bounds [19]. Task-specific finite-rate optima also arise in nuclear counting; for example, Bécares and Blázquez optimized counting rate by balancing dead-time losses, counting uncertainty, source multiplicity, and dead-time-calibration uncertainty [14]. Grönberg et al. derived an exact finite-frame count probability mass function for a seminonparalyzable detector with nonzero pulse length, validated its moments against an electronic Monte Carlo model, and used the exact distribution together with Poisson and Gaussian approximations to evaluate Fisher information for a single-threshold soft-tissue-density task as a function of count rate; their fixed-frame Fisher curve exhibits an interior maximum followed by high-rate decline [20].

A complementary line retains richer event information. Rapp et al. modeled the sequence of dead-time-distorted detection times as a Markov chain, enabling accurate high-flux single-photon lidar and large acquisition-time reductions [8]. Jorgensen and Johnson derived asymptotic Fisher-information rates and efficient estimators for periodic dead-time event-detection processes, showing that activation and event statistics can retain information discarded by conventional histogram summaries [12].

Here we isolate a different detector-channel question: an ideal nonparalyzable detector is active at the start of a finite exposure, only its integrated scalar count is retained, and the parameter of interest is the log arrival rate itself. Treating $\rho = \lambda\tau$ and $\nu = T/\tau$ as independent axes, we derive the exact finite-window information surface, the terminal-phase missing-information identity of Eq. (3), and the principal information ridge of Eq. (5). These statements concern the active-start ideal count-only channel and do not replace task-specific finite-rate optima for nonzero-pulse-length detectors or information bounds for time-resolved dead-time event records.

5 Study 1: dense patterns — the multiplex-limited regime

5.1 Preregistered design

We tested whether renewal quasi-likelihood reconstruction permits operation at $\bar{\rho} = 0.6$ with a statistically significant reduction in total optical integration time relative to the conventional $\bar{\rho} = 0.05$ operating point, on 64^2 scenes under dense Bernoulli illumination (occupancy 50%). Scenes are the units of inference: 24 confirmatory detail targets (six structural families $\times$ four instances), each measured at nine dwell points $\nu \in \{5, \dots, 2000\}$ with five measurement seeds. Per scene, seed-mean radiometric PSNR versus optical integration time is isotonicity regressed for each arm; the quality target is Q_{90} , the safe-arm minimum plus 90% of the safe-arm span, and the acquisition-speed ratio S_j is the ratio of safe to high-flux crossing times, with a frozen censoring taxonomy (a safe-arm span below 0.5 dB is uninformative; a high-flux arm that never reaches Q_{90} scores $S_j = 0$). The confirmatory gate — median $S \geq 3$, family-stratified nested-bootstrap (10 000 replicates) lower bound > 1 , and $\geq 18/24$ scenes with $S_j > 1$ — and every constant above were frozen in an immutable commit before any confirmatory reconstruction existed, together with a fixed-dwell secondary endpoint (median ≥ 1 dB at the terminal dwell, lower bound > 0 , $\geq 18/24$ positive).

5.2 Preregistered outcome

The confirmatory time-to-quality gate returned negative: in 24/24 scenes the safe-arm quality span across the entire dwell grid fell below the preregistered 0.5 dB informativeness floor (SAFE_RANGE_UNINFORMATIVE; $S_{\text{gate}} = 1$ by the frozen censoring taxonomy), so no time-to-quality acceleration could be certified. The preregistered fixed-dwell secondary, in contrast, was uniformly positive but small: at $\nu = 2000$ the high-flux arm gained a median +0.28 dB (radiometric PSNR), positive in 24/24 scenes, with family-stratified bootstrap lower bound +0.23 dB — statistically significant, but far below the preregistered 1 dB practical-relevance bar. Both outcomes are reported as frozen; no endpoint, gate, or analysis constant was altered after the freeze.

5.3 Natural-scene cohort (secondary, no gate)

On the preregistered natural-image cohort (30 scenes, same dense-pattern protocol, no confirmatory gate), the acquisition-speed endpoint was safe-range-uninformative for 12/30 scenes, while the fixed-dwell gain was larger and more uniform than on the detail

cohort: median +1.44 dB, 29/30 scenes positive, family-stratified bootstrap lower bound +1.18 dB. For context, classical correlation ghost imaging on the identical measurement records scores 4.8–6.3 dB — at this photon budget (≈ 95 detected photons per pixel over the full acquisition) the likelihood-based reconstruction, not the sampling ratio, is what renders the data usable. Smooth, prior-limited scenes thus occupy the low-benefit region of the acquisition-speed map even where the fixed-dwell radiometric benefit of high-flux operation is substantial.

5.4 Mechanism: bucket multiplex contrast

The two outcomes are jointly explained by the bucket contrast of dense multiplexed illumination. For occupancy-1/2 patterns on n pixels the bucket energies $u_i = a_i^\top x$ concentrate as $C_u = \text{sd}(u)/\text{mean}(u) \sim 1/\sqrt{n}$ ($\approx 1.6\%$ at 64^2): the structural information rides on a relative modulation of order 10^{-2} , so the per-count information about *structure* is contrast-limited, not count-limited — raising the flux adds photons to a nearly constant bucket. This is the multiplex-limited regime of the operating map; it motivates Study 2.

6 Study 2: the contrast ladder — equal load, equal dose

6.1 Preregistered design

Study 2 varies spatial occupancy $k \in \{512, 32, 16, 1\}$ (occupancy 50%, 3.1%, 1.6%, 0.1% of $n = 1024$ pixels at 32^2) using binary matrices with exact row and column sums k , while holding the *mean detector load and the incident photon dose fixed across the ladder*: the reconstruction operator is $A^{(k)} = (n/k) B^{(k)}$ at flux $\Phi = \bar{\rho}/\tau$, so every occupancy shares identical mean count rate, identical incident photons per pattern, and identical cumulative dose per object pixel; only the on-pixel peak intensity rises as n/k . Sparse patterns therefore cannot win by starving the detector — the mechanism prior sparse-pattern work uses to mitigate pile-up [15, 16]. Only the $k = 16$ arm carries the confirmatory gate (constants unchanged from Study 1); $k = 32$ is a preregistered robustness arm and $k = 512$ and $k = 1$ are controls, all without gates. Two mechanism variables are recorded descriptively and never enter any gate: the measured bucket-energy contrast $C_u = \text{sd}_i(u_i)/\text{mean}_i(u_i)$ and the contrast-to-count-noise index $\Gamma = C_u \sqrt{\nu\rho/(1+\rho)}$, whose crossing of unity marks the onset of the photon-limited structural regime. A contrast-to-noise ratio on generator-frozen signal/background regions is carried as a no-gate co-secondary, and all photon budgets are reported on two

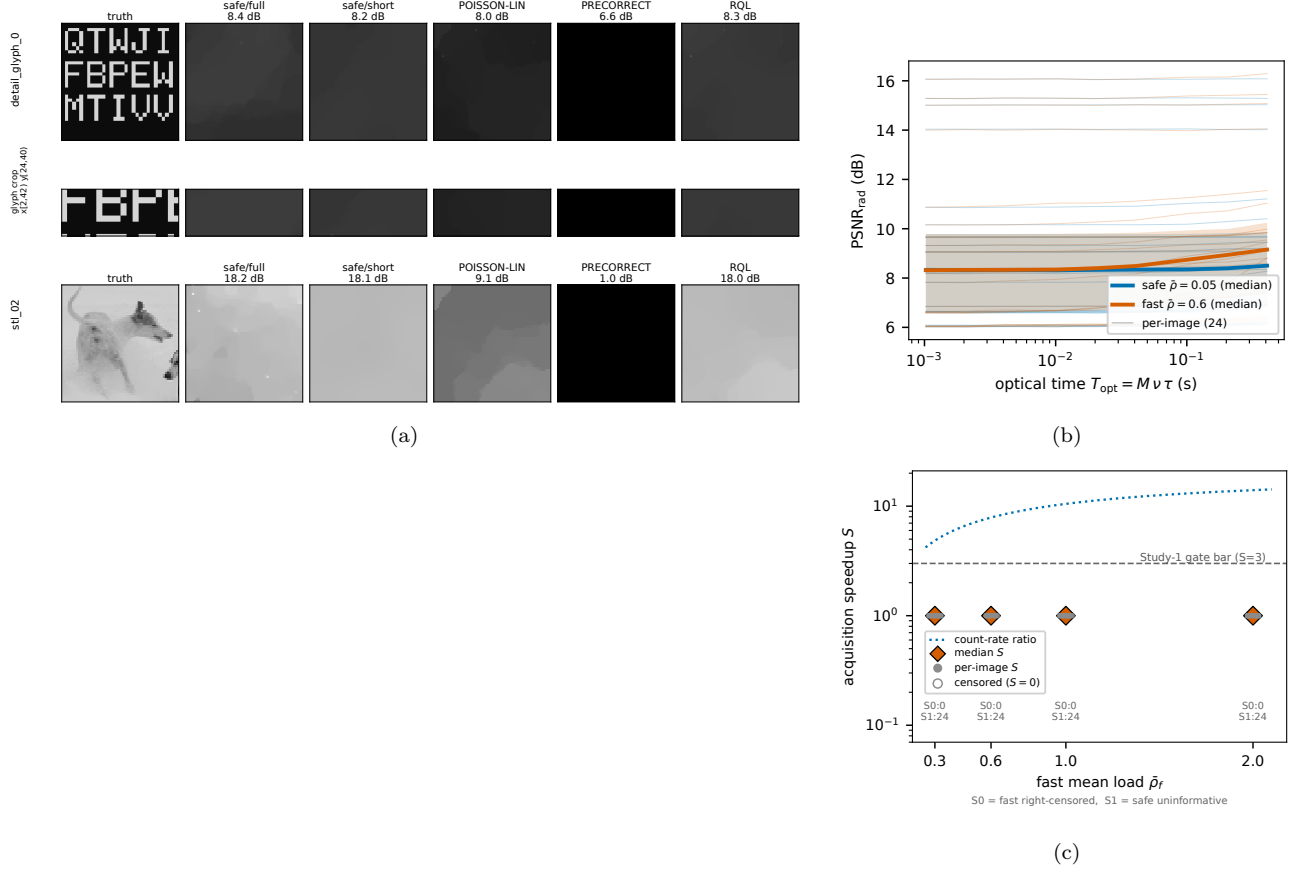


Figure 2: **Dense illumination leaves the preregistered speed endpoint uninformative.** (a) Detail and natural-scene boundary examples under safe and high-flux reconstructions. Dense multiplexing leaves the detail target prior-limited, whereas RQL retains coarse structure in the natural example. (b) Radiometric PSNR versus optical integration time: the safe-arm quality range remains below the frozen informativeness floor. (c) Every detail scene lies at the $S = 1$ censoring bound despite the available detector count-rate increase, exposing the multiplex limitation.

axes: the measured-photons axis $S_{det} = (Mn)^{-1} \sum_i N_i$ (the convention of Ref. [4], whose experiments span $S_{det} = 0.096\text{--}0.44$) and the incident axis S_{inc} , which differ nonlinearly under dead time.

6.2 Preregistered verdicts

The preregistered Study-2 primary endpoint was formally negative because its breadth criterion failed: 16 of 24 scenes accelerated, below the required 18 of 24. The two effect-size criteria passed, with median $S_{gate} = 6.796$ and family-stratified bootstrap lower bound 2.835, but they do not rescue the failed primary. The preregistered fixed-dwell secondary passed: median +1.16 dB at the terminal dwell, 19 of 24 scenes positive, and lower bound +0.72 dB. The frozen hierarchy is therefore “primary formally negative; fixed-dwell secondary passed,” with no endpoint, geometry, or image-class redesign.

The eight non-accelerating scenes comprise the en-

tire microtexture family (0/4), three of four contour instances, and one chirp instance, whereas glyph, maze, and spokes accelerated 4/4. After unblinding, a descriptive family-level analysis showed that the six success fractions were monotonically ordered by their mean measured bucket contrasts: maze $C_u = 0.53$ (4/4), glyph 0.41 (4/4), spokes 0.38 (4/4), chirp 0.28 (3/4), contour 0.27 (1/4), and microtexture 0.17 (0/4). This ordering was not preregistered, did not enter any decision rule, and does not estimate a universal critical contrast.

Across the preregistered no-gate occupancy controls, fixed-dwell gain was nonmonotone and largest among the tested rungs at $k = 32$ (3.1% occupancy), where the median was +4.13 dB. This is evidence of a contrast-conditioning trade-off, not an “optimal sparsity” claim. Complete per-rung and per-scene values are given in the supplement.

6.3 Contrast and conditioning across occupancies

The measured bucket contrast follows the designed ladder: $C_u = 0.040, 0.222, 0.316$, and 1.270 at $k = 512, 32, 16$, and 1 , respectively. At $\bar{\rho} = 0.6$ and $\nu = 2000$, the corresponding descriptive contrast-to-count-noise index is $\Gamma = 1.09, 6.07, 8.66$, and 34.79 (Fig. 3). Sparsity increases bucket modulation but also raises the conditioning burden $\kappa_A = n/k$; the competing trends explain why fixed-dwell benefit is nonmonotone rather than increasing steadily with contrast. The dense-rung value remains close to the Study-1 magnitude, showing that the change from 64^2 to 32^2 did not manufacture the effect.

6.4 Mechanism separation: equal-load gain is not rate starvation

On the six frozen mechanism representatives we ran the $k = 16$ geometry in two flux conventions: (i) *normalized* — the Study-2 mechanism, equal mean load and equal dose; and (ii) *unnormalized* — sparsity is allowed to reduce the detector count rate, as in prior sparse-pattern mitigation. In the normalized convention the high-flux arm retains a positive gain of $+1.81$ and $+1.10$ dB at $\nu = 200$ and 2000 , respectively, at detector loads of $S_{\text{det}} = 0.072$ – 0.72 counts per pattern per pixel. The unnormalized convention shows larger *relative* deltas but strictly lower absolute quality at every dwell (e.g. 7.45 versus 10.26 dB at the safe point, $\nu = 2000$): it wins its comparison by starving the detector. The equal-load, equal-dose gain is therefore a contrast effect, not a rate-reduction effect.

6.5 Dynamic scattering

Under multiplicative dynamic scattering (lognormal AR(1) factors, $\text{CV}(\alpha) = 0.20$, applied before dead time) the high-flux gain persists and grows with dwell: $+0.66$, $+1.84$, and $+2.32$ dB at $\nu = 20, 200$, and 2000 , respectively (Supplementary Information), reported on the measured-photons axis S_{det} for direct comparison with single-photon underwater single-pixel experiments [4].

6.6 Robustness to detector mismatch

Detector-mismatch consequences were quantified descriptively on the 64^2 cohort at the preregistered anchor ($M/n = 0.5$, $\nu = 500$), pairing each perturbed cell with its no-mismatch baseline. Every anchor perturbation leaves the RQL radiometric PSNR within 0.34 dB of baseline at both operating points: dead-time miscalibration $\hat{\tau} = \tau(1 \pm 0.1)$ costs at most 0.12 dB (with a $\pm 6\%$ radiometric flux bias, the largest observed); a 10% dark

rate costs 0.16 dB when calibrated and 0.34 dB when unmodeled; afterpulsing at 2–5% costs at most 0.09 dB. Under the same perturbations the pre-correction arm degrades more in five of six anchors (up to 0.79 dB, unmodeled dark). Start-mode, guard-time, and dead-time-jitter axes, severity sweeps to 50% dark, and the descriptive measurement-audit distributions are tabulated in the supplement.

6.7 Exact-likelihood reference at small scale

Because RQL is a convex surrogate, we verified at 8^2 and 16^2 that it tracks the exact renewal likelihood under an identical TV functional and the identical frozen selection rule: across the full preregistered grid ($\rho \in \{0.03, \dots, 2\}$, $\nu \in \{20, \dots, 2000\}$, three seeds, six scenes, 864 paired reconstructions) the median exact-minus-RQL radiometric-PSNR gap is ≤ 0.01 dB throughout the deployment zone and reaches only -0.05 dB in the extreme-load corner (16^2 , $\rho \geq 0.6$, $\nu = 2000$). Individual cells differ in both directions (up to 9 dB), consistent with the nonconvexity of the exact objective, with no systematic advantage to either arm; full tables and multistart diagnostics are provided in the supplement. The convex surrogate therefore tracked the exact reference closely on this grid. Under the frozen TV settings, the exact and approximate likelihood reconstructions agreed to the stated numerical tolerance, and the exact calculation added little runtime at those tested sizes. This empirical agreement does not establish uniqueness of the TV-regularized solution; convexity of the data term alone is insufficient when the design is underdetermined and the TV penalty is not strictly convex.

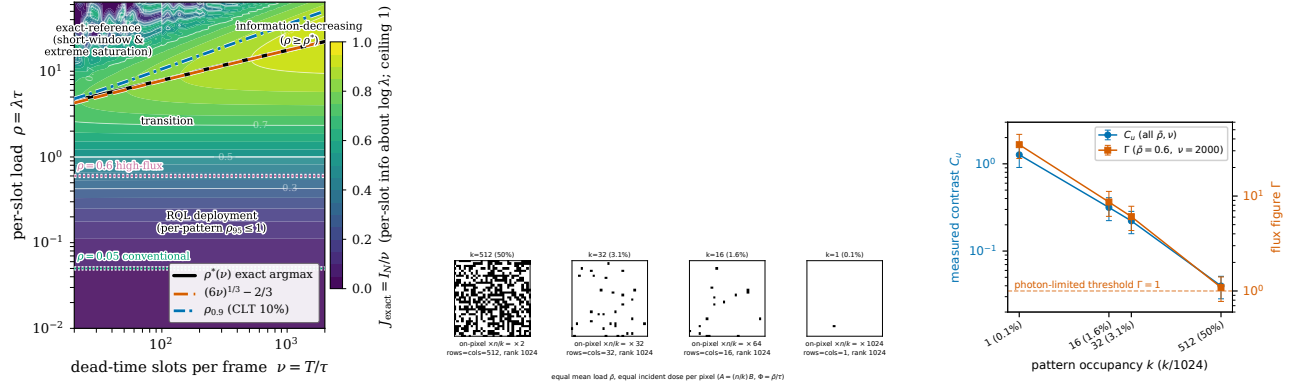


Figure 3: **Exact information ridge and the illumination ladder.** (a) Exact count-Fisher information per unit time versus detector load, peaking on $\rho^*(\nu) = (6\nu)^{1/3} - 2/3$. (b) Representative masks at $k \in \{512, 32, 16, 1\}$; amplitude normalization holds mean load and incident dose per pixel fixed across occupancies. (c) Measured bucket contrast C_u and descriptive photon-limited index Γ versus occupancy. Contrast rises with sparsity, while the dense rung lies near the descriptive $\Gamma = 1$ onset proxy. Error bars are ± 1 s.d. over image-seed pairs.

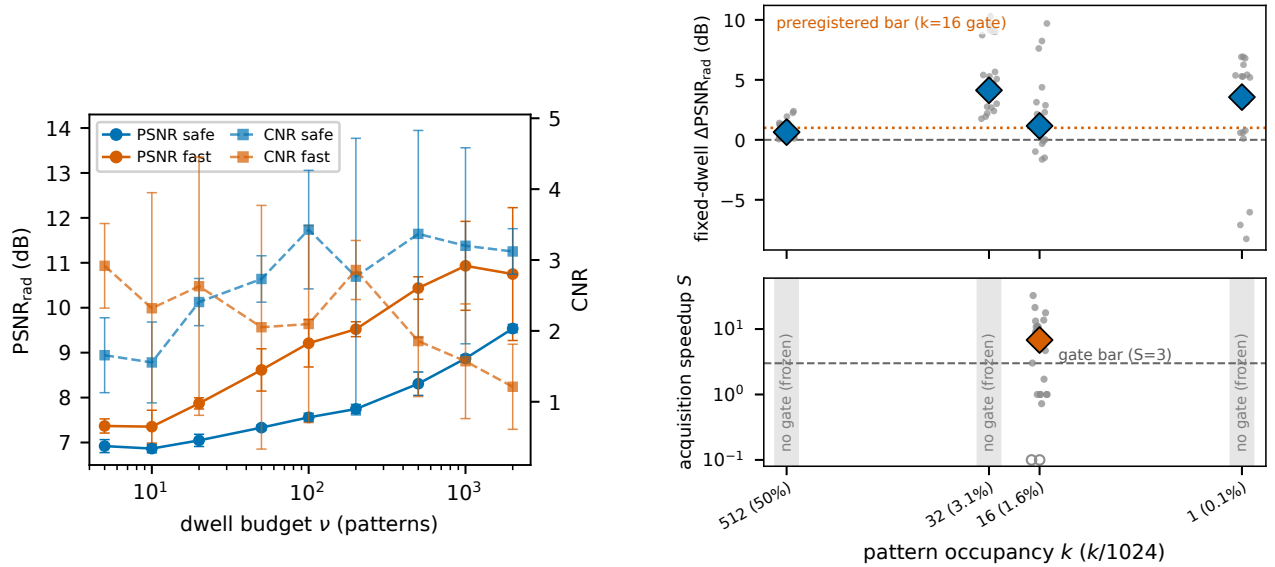


Figure 4: **Dwell curves, fixed-dwell gain, and acquisition speed.** (a) Radiometric PSNR and CNR versus dwell budget for the preregistered subject at safe and high flux. (b) Per-image fixed-dwell gain across the occupancy ladder and per-scene acquisition speed at the gate-carrying $k = 16$ rung. The Study-2 primary is formally negative because only 16 of 24 scenes accelerate, despite median $S_{\text{gate}} = 6.796$; the fixed-dwell secondary passes at +1.16 dB. Open markers denote censored scenes. Only $k = 16$ carries the frozen speed gate.

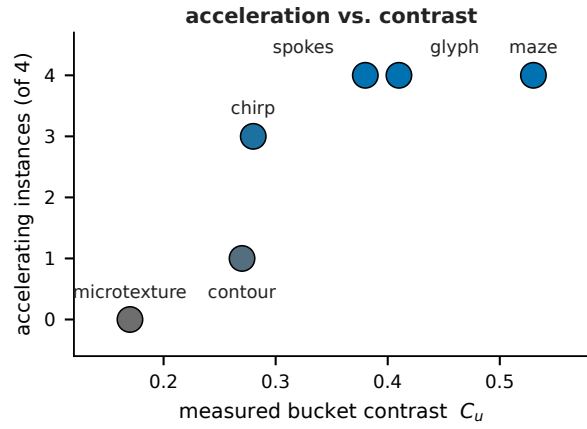


Figure 5: **High-flux acceleration by scene family (empirical)**. Acceleration rises with measured bucket contrast across the six structural families: maze, glyph, and spokes accelerate 4/4, chirp 3/4, contour 1/4, and microtexture 0/4. These outcomes use one operating load; no critical-contrast boundary or transition band is claimed.

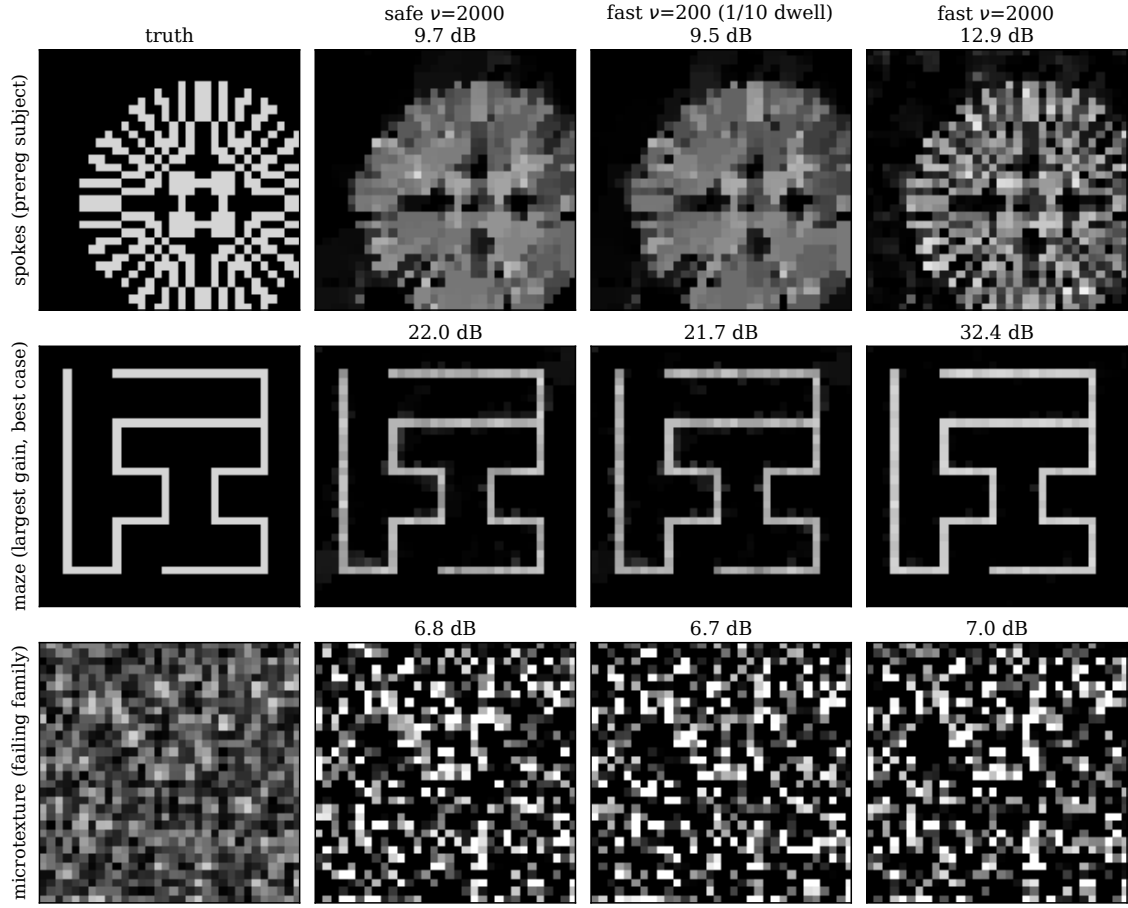


Figure 6: **Study-2 reconstruction outcomes at the gate-carrying occupancy $k = 16$.** Columns show truth, the safe operating point at full dwell, high flux at one tenth of the dwell, and high flux at full dwell. Contrast-adequate spokes and maze scenes reach the safe quality sooner and improve at equal dwell, whereas low-contrast microtexture remains insensitive to flux. The rows comprise the preregistered subject, a clear success, and a failing-family example; detailed selection provenance is in the supplement.

7 Discussion

The scientific position established by the two studies is deliberately mixed. Dense illumination is multiplex-limited; sparse equal-load illumination produces large and statistically supported benefits for many, but not sufficiently many, preregistered scenes to pass the breadth gate. The scene response is nevertheless structured by bucket contrast, while the tested occupancy ladder exposes an independent conditioning penalty. Together with the exact count-information ridge and the passed fixed-dwell secondary, this constitutes an operating map: a computable answer to when high-flux operation is worth engaging, rather than a universal acceleration claim.

7.1 Limitations

Three scope boundaries must be stated plainly. First, all results are simulation-based; the exact renewal forward model, detector-mismatch envelopes, and frozen benchmark do not substitute for optical-bench validation. Second, sparse operation at equal dose concentrates the same illumination into fewer pixels: the on-pixel peak irradiance rises as n/k ($64\times$ at the gate-carrying occupancy), so source power, spatial-light-modulator limits, laser safety, or local photodamage may bind before the detector does. Third, the operating map assumes a calibrated, stable detector dead time. Jitter, drift, correlation, or paralyzable recovery changes the count-information surface — the companion manuscript [21] derives the hold-time-jitter cap on the optimal load and its global power-control deployment; deployment therefore requires recovery-time calibration and a detector-specific load limit. The reported sparse rungs are conditional on both peak-power headroom and that calibration.

Three extensions follow directly from the failure structure. For uniformly scattered fixed- k supports, $C_u^2 = \frac{n-k}{k(n-1)} (n\|x\|_2^2 - 1)$, giving the observed $k^{-1/2}$ spatial-averaging law. A local support instead averages only the effective number of independent correlation cells within the patch and may therefore preserve modulation from scale-matched structure. Local, foveated and multiscale single-pixel patterns have previously been developed for resolution and measurement efficiency; optimizing their spatial scale for equal-load dead-time contrast is a distinct future design problem.

The present results do not establish that adaptive illumination is necessary. Although dead-time nonlinearity makes the statistically efficient pattern scale scene-dependent, a strict adaptive advantage requires a specified scene class, resource constraint and loss. A natural follow-up is to seek a minimax or Chernoff-information separation between fixed multiscale illumination and

a two-stage scale-estimation-plus-matched-pattern policy.

A coherent extension would use a coarse complex-field estimate to synthesize an optical displacement that suppresses the predicted common mode before photon counting. Unlike electronic balanced subtraction, a dark-port receiver removes photons before dead-time loss. Exact nulling, however, gives quadratic residual sensitivity and demands spatial-mode matching and phase stability; the relevant operating variable is therefore an optimized displacement bias rather than maximal null depth. Testing this receiver first under static and then dynamic scattering is a natural optical-bench extension. More broadly, architectures that retain per-photon timing [7, 8] occupy a different information regime than the count-only channel analyzed here; the map presented in this work is the count-only baseline against which such richer architectures can be costed. The proposed operating map, the deployment-legal reconstruction rule, and the frozen open benchmark open up an avenue for dwell-time-limited and photon-budget-limited single-pixel systems operating deep in the dead-time regime.

Code and data availability

All simulation code, frozen preregistration documents, per-cell manifests, and evidence bundles are available at [\[repo URL wording — user decision\]](#).

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[\[funding/acknowledgments — user decision\]](#)

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